Charge + light foundation model (qlfm): status

Data/sim width check, new MicroBooNE and SBND samples, first real-data test

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fm4npp-lartpc ▪ issues #8–#11

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Prepared with the assistance of Claude (Anthropic). Code and numbers: `https://github.com/HaiwangYu/fm4npp-lartpc`.

Summary

1. **Sunny's data/sim width check (#9).** qlfm-M CL relies on the simulated hit width: the measured data/sim rescale costs -0.21 in hit F1 (T1). Retraining at data-like widths keeps most of the score *in simulation*; removing width costs about 0.09.

2. **MicroBooNE real-data samples (#10).** Round 1 is delivered: BNB and ν_e overlay, beam-on, EXT, each about 1k events with Wire-Cell labels. The processing matches the open data exactly. Clean pools for round 2 are identified.

3. **SBND samples (#11).** The reco1 + Wire-Cell $\rightarrow$ HDF5 chain works; the 10k BNB sample converts and trains. MicroBooNE pretraining transfers (T1 0.85 vs 0.60–0.70 from scratch). The fixed Wire-Cell labels still lead on T1 (0.91).

4. **First real-data check (#10).** In simulation (overlay), qlfm beats Wire-Cell and Pandora. **On real beam-on data Wire-Cell wins**: qlfm-M CL finds only about 20% of Wire-Cell's ν charge (90% in simulation). The size of the drop follows each model's reliance on hit width.

1a. Sunny's finding, and our test: perturb simulated ν hits, no retraining (#9)

Sunny (6 Nov). In the overlay, every ν is simulated and every negative is a real cosmic.

- Simulated hits broaden faster with drift: data/sim width $\approx$ 0.92–0.94 at the cathode (Y). The simulation is over-diffused: D_L 6.40 vs 3.74 cm^2/s.
- Per hit, width barely separates data from sim (AUC 0.51–0.58), but a model that pools many hits might still use it.

Our test (`eval_perturb.py`): rescale the simulated ν hits' width (2D factors, per plane and drift) or charge ($\pm$2, 5%), then re-evaluate T1 and T2.

Δ vs nominal. Multi-task fine-tune on 1% of labels, mean of 3 seeds.

model	task	nominal	sim width $\times$ meas.	all widths $\times$ meas.	charge $\pm$5%
qlfm-M CL	T1 F1	0.919	-0.212	-0.191	$\leq$0.001
	T2 AUC	0.993	-0.034	-0.025	$\leq$0.001
qlfm-M C+XF	T1 F1	0.859	-0.029	-0.008	$\leq$0.001
	T2 AUC	0.990	-0.003	0.000	$\leq$0.001
PTv3+XF	T1 F1	0.840	0 (sees no width)	0	0.001

Test-set 95% half-width: 0.001–0.003.

- **Charge shift:** negligible for every model.
- **Width:** 50–100$\times$ the test uncertainty for qlfm-M CL. Rescaling *all* hits (sim/real ratio unchanged) hurts almost as much, and so does broadening: the model has learned a sharp width-vs-drift relation.
- **Reading:** with light, the model knows t_0, so diffusion width tells drift distance. That is real physics, learned on an over-diffused simulation.

1b. Fix (b): fine-tune with data-like or no width (#9)

Test, 1% labels, fine-tuned, 3 seeds. Width change applied in training *and* test.

model	widths	T1↑	T2↑	T3↑	T4↓	T5↓
qlfm-M CL	nominal	**0.922**	0.988	0.748	0.225	0.529
	data-like	**0.914**	0.993	0.746	0.227	0.528
	removed	0.829	0.968	0.685	0.246	0.601
qlfm-M C+XF	nominal	0.862	0.969	0.685	0.283	0.585
	data-like	0.831	0.963	0.673	0.286	0.597
	removed	0.809	0.953	0.660	0.288	0.610
PTv3+XF (1 seed)	n/a	0.831	0.976	0.789	0.227	0.592

T1 per-event median F1; T2 AUC on bright-flash events; T3 flavour macro-F1; T4 E_ν resolution; T5 PMT-light MSE.

- **In simulation,** retraining at data-like widths keeps qlfm-M CL's T1 lead (0.914).
- **Without width,** the lead disappears (0.829, PTv3's level). The T1 advantage *is* width.
- **Frozen M CL** is the most width-dependent: T1 0.894 / 0.847 / 0.533 (nominal / data-like / removed).
- **New:** Sunny's *3D* per-plane factors (U 0.88, V 0.94, Y 0.90 at full drift) arrived. Not applied yet.
- **Open:** these are still simulation scores. The real-data test is section 4.

2a. MicroBooNE round-1 samples with Wire-Cell labels (#10)

Made at FNAL with our own HDF5Maker build (open-data layout plus a real-data mode). Wire-Cell hit labels come from the WC nusel files.

sample	events	WC-labelled	WC generic ν	hits
BNB overlay (MCC9.1 run1)	1013	959	305 (32%)	15.6M
ν_e overlay (intrinsic)	1271	1271	865 (68%)	21.7M
beam-on (run1 C1)	1041	849	71 (8.2%)	17.2M
EXT (run1 C2)	1248	1248	27 (2.2%)	19.8M

WC ν hit label = `wc_main` (in-time main cluster) & `wc_generic_nu`. Against truth in the overlay: efficiency 0.53 / 0.52, purity 0.95 / 0.97 (BNB / ν_e).

Checks on Polaris

- The same converter handles all four files; a real-data mode covers the absent truth.
- **Identical** to the published open data on the 849 shared BNB events: hits, light, truth. Light is `*Stage1Optical` in both data and MC.
- **Overlap:** 783 / 1058 of the BNB / ν_e overlay events are in our *train* split. Clean overlay events: 230 BNB, 213 ν_e. Beam-on and EXT are fully clean.

2b. Round 2: clean candidate pools (not started; needs approval)

An event is *clean* for a qlfm-vs-Wire-Cell test if it is not in our train split (train = pretraining set and source of fine-tune labels).

BNB overlay

- 155,337 events are in the WC run1 checkout but *not* in the published HDF5: 4,595 public artroot files, including 2,196 files (98k events) where every event qualifies.
- About 3k events ≈ 70 artroot files plus about 13 GB of nusel staging.
- Fallback: our test (38.8k) or val (38.0k) split.

ν_e overlay

- Every public ν_e event is in the published HDF5, but our corpus never used two of its files, `nue_NoWire_01` and `_04` (≈12k events).
- Fallback: our test (3.0k) or val (3.9k) split.

Real data

- More beam-on and EXT is the most valuable addition (slide 4b).

Event lists for FNAL: `fm4npp-lartpc-data/uboone-round1/requests/`.

3a. SBND sample production on Aurora (#11)

Chain (`sbnd-h5maker`): Gen2 reco1 MC (sbndcode v10_14, CORSIKA cosmics) plus Wire-Cell aurora-r2 output → MicroBooNE open-data HDF5 layout.

- **Per-hit truth:** reco1 has no hit back-tracking, so 3D G4 deposits are projected onto hits (SCE, drift, channel, tick). 2.4% of hits get no deposit.
- **Per-ν truth:** two ν in a spill happen often.
- **Reco baseline:** the Wire-Cell ν-cluster footprint per hit (no Pandora here).
- **Bug found in review:** WC numbers channels plane-major, LArSoft TPC-major, so only 2 of 6 planes matched. Fixed in v0.4: label efficiency 0.29 → 0.85.
- **qlfm:** `QLFM_DETECTOR=sbnd`. Two TPCs (TPC 1 mirrored onto one drift axis), 120 PMTs, beam window 0–3 μs. MicroBooNE checkpoints load; only the PMT embeddings start fresh.

10k BNB CV	
events / files	10,754 / 21
ν / vertex in AV	19,402 / 4,807
hits (per event)	132M (12.3k)
ν / cosmic hits	6.4M / 123M
WC ν hits: eff. / purity	0.66 / 0.90
cost	0.35 node-h, 3.9 GB
full aurora-r2 (not run)	
BNB CV	627k
ν_e CC	269k
NC Δ radiative	27k
estimate	12–30 node-h, 0.33 TB

3b. SBND 10k fine-tune test: MicroBooNE pretraining transfers (#11)

Test split (≈1k events; 425 for T1). All training labels, 5 tasks jointly, 3,000 steps (17 min on one A100).

model	T1 F1↑	T2 AUC↑	T3* ↑	T4↓	T5↓
qlfm-M CL, MicroBooNE-pretrained, fine-tune (2 seeds)	**0.852 / 0.851**	0.994	0.80 / 0.85	**0.24**	**0.66**
qlfm-M C+XF, MicroBooNE-pretrained	0.791	0.989	0.76	0.27	0.74
qlfm-M CL, frozen	0.486	0.977	0.71	0.28	0.84
same model, from scratch (2 seeds)	0.600 / 0.699	0.984 / 0.989	0.74 / 0.76	0.26 / 0.27	0.97
Wire-Cell ν clusters (fixed labels)	**0.912**				

T1 charge-weighted F1. *T3: ν_μCC vs NC macro-F1 (only 4 ν_eCC in test).

- **Transfer works:** pretrained beats scratch on every task and is stable over seeds. T5 improves too, though the 120 PMT embeddings are new.
- **Frozen** is poor on T1: the features need adapting to SBND.
- **Wire-Cell leads T1 by 0.06**, but the runs were short: 3.6k hit-labelled events, still improving.
- **Needed:** a ν_eCC sample (BNB CV has 30 per 10k). Full production awaits approval.

4a. Overlay (simulated ν + real cosmics): qlfm beats Wire-Cell and Pandora (#10)

T1 charge-weighted F1, events with a visible in-AV ν (all / clean)

	BNB 379 / 71 events	ν_e 1254 / 211 events
qlfm-M CL, nominal widths (3 seeds)	**0.92 / 0.93**	**0.94 / 0.93**
qlfm-M CL, data-like widths (3 seeds)	0.91 / 0.93	0.93 / 0.93
qlfm-M CL, width removed	0.83 / 0.84	0.90 / 0.90
PTv3+XF (3 seeds)	0.82 / 0.81	0.91 / 0.91
WC main cluster	0.88 / 0.92	0.85 / 0.83
WC main & generic ν (the label)	0.82 / 0.88	0.73 / 0.63
Pandora (max ν-score slice)	0.64 / 0.68	0.76 / 0.76

T2, ν vs no-ν (BNB + ν_e overlay)

	all 1633 / 189	clean 282 / 37
qlfm-M CL AUC	0.993–0.997	0.994–0.999
PTv3+XF AUC	0.995	0.990
Pandora AUC	0.85	0.89
WC generic ν: efficiency	0.70	0.61
at false-positive rate	1.1%	0%
qlfm-M CL efficiency at WC's FPR	**0.95–0.97**	**0.96–0.99**

Existing 1%-label multi-task models, no retraining. "Clean" = not in our train split.

4b. Real beam-on data: Wire-Cell wins, and the drop follows hit width (#10)

model	share of WC's ν charge that the model also tags: overlay (simulated ν)	**beam-on (real ν)**	beam-on pass rate with EXT fixed at 2.2%
Wire-Cell generic ν	–	–	**8.2%**
PTv3+XF (sees no width, 3 seeds)	0.76–0.78	0.56–0.58	7.2–7.8%
qlfm-M C+XF	0.82	0.60	5.7%
qlfm-M CL, width removed	0.80	0.63	4.9%
qlfm-M CL, data-like widths (3 seeds)	0.89	0.35–0.39	4.9–6.0%
qlfm-M CL, nominal widths (3 seeds)	0.89–0.90	**0.18–0.24**	3.9–5.4%

849 beam-on and 1248 EXT events with WC labels. Hit agreement on the 70 (beam-on) / 305 (overlay) WC-selected events. Pass rate: model cut set so EXT passes at WC's rate; binomial error $\approx \pm 0.9\%$.

- **qlfm-M CL nominal** calls the same charge share ν in beam-on as in EXT (0.6% vs 0.5%): it finds essentially no real neutrinos.
- **Drop $\propto$ reliance on width.** Data-like 2D widths help only partly; removing width leaves a smaller, uniform gap (0.80 $\rightarrow$ 0.63).
- **Wire-Cell** gives the largest beam-on excess at fixed EXT rate. Its tuning carries over to data better than our heads do.
- **Caveats:** on data, WC is a reference, not truth (purity $\approx$0.96 in the overlay). Small samples; no POT scaling.

For discussion: next steps (tracked in #12)

Closing the data/sim gap

- Sunny's 3D per-plane width factors in training: fine-tune rerun (≈3 node-h) or full re-pretraining.
- Width randomisation or dropout, or no width input at all.
- Wire-Cell as labeller: fine-tune on real data (beam-on/EXT) with Wire-Cell ν labels.
- Pretrain on data-like sim, freeze the backbone, fine-tune on nominal (sim-like) sim.
- Pretrain on real data, freeze the backbone, fine-tune on data-like sim.

Samples

- MicroBooNE round 2: more beam-on/EXT plus clean overlay.
- SBND: ν_eCC 10k (0.35 node-h) and the full 923k (12–30 node-h).

SBND modelling

- Longer fine-tune: can qlfm close the 0.06 T1 gap to Wire-Cell?
- SBND pretraining once the full sample exists.
- The data/sim question applies to SBND too (all-simulation sample, no overlay).

Paper (ME1, #8)

- The T1/T2 "light helps" claim must hold at data-like widths *and* on real data.
- Should the headline evaluation be on real data?
- Fair comparison: give hit width to the other models (PTv3, …), or remove width from all of them.